

Y. Datu Adiatma, Ph.D.

Postdoctoral Research Associate

Department of Earth Ocean and Atmospheric Science

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Education

PhD in Earth Science, The Ohio State University Fall 2023

Dissertation title: Chemostratigraphic and numerical modeling constraints on Ordovician climate, carbon cycling, and carbonate diagenesis

Advisor: *Matthew R. Saltzman*

MS in Earth Science, The Ohio State University Fall 2018

Thesis title: Did early land plants produce a step-change in atmospheric oxygen centered on the Late Ordovician Sandbian Age ~458 Ma?

Advisor: *Matthew R. Saltzman*

BS in Geology, Institut Teknologi Bandung Spring 2014

Thesis title: The geology and shale gas reservoir potential of Tanjung Formation in Suato District, Barito Basin, Indonesia

Advisors: *Benyamin Sapiie* and *Dwiharso Nugroho*

Academic Appointments and Research Experience

Postdoctoral Research Associate, Florida State University 2024 - present

Conducted field campaigns to collect samples and perform geochemical analyses (thallium isotopes, trace metal concentrations, Fe speciation, I/Ca) on Paleozoic-aged sedimentary rocks to reconstruct biogeochemical cycling and the paleoredox states of Cambrian - Ordovician oceans, and mentor undergraduate and graduate students.

Postdoctoral Research Associate, The Ohio State University Fall 2023

Performed calcium isotope analyses ($\delta^{44/40}\text{Ca}$) on Paleozoic-aged carbonate rocks and developed numerical models to constrain the role of diagenesis in affecting geochemical proxies in carbonate rocks.

Graduate Research Associate, The Ohio State University 2019 - 2023

Performed laboratory maintenance (e.g., repairing Milli-Q system, replacing fume hood motors, and scroll pump maintenance), assisted visiting scientists in the Clean Lab and TIMS Lab, separated conodont apatite fossil materials using LMT heavy liquid, collected samples, performed geochemical analyses ($\delta^7\text{Li}$, $^{87}\text{Sr}/^{86}\text{Sr}$, ϵ_{Nd} , Sr/Ca , $\delta^{44/40}\text{Ca}$) on Ordovician-aged carbonate rocks, and developed a suite of numerical models to reconstruct changes in global silicate weathering and its role in causing climate cooling.

Publications

Peer-reviewed articles

- 2024 **Adiatma, Y.D.**, Saltzman, M.R., Liu X-M., Wang, X-K., Edwards, C.T., Lithium isotope stratigraphy and Ordovician weathering. *Earth and Planetary Science Letters* 647, 119030.
- 2024 **Adiatma, Y.D.**, Saltzman, M.R., Griffith, E.M., 2024. Calcium isotope constraints on a Middle Ordovician carbon isotope excursion. *Earth and Planetary Science Letters* 641, 118805.
- 2022 Avila, T.D., Saltzman, M.R., **Adiatma, Y.D.**, Joachimski, M.M., Griffith, E.M., Olesik, J.W., 2022. Role of seafloor production versus continental basalt weathering in Middle to Late Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$ and climate. *Earth and Planetary Science Letters* 593, 117641.
- 2022 Conwell, C.T., Saltzman, M.R., Edwards, C.T., Griffith, E.M., **Adiatma, Y.D.**, 2022. Nd isotopic evidence for enhanced mafic weathering leading to Ordovician cooling. *Geology* 50, 886-890.
- 2019 **Adiatma, Y.D.**, Saltzman, M.R., Young, S.A., Griffith, E.M., Kozik, N.P., Edwards, C.T., Leslie, S.A., Bancroft, A.M., 2019. Did early land plants produce a stepwise change in atmospheric oxygen during the Late Ordovician (Sandbian ~458 Ma)? *Palaeogeography, Palaeoclimatology, Palaeoecology* 534, 109341.

Manuscripts in review / in preparation

- In review* **Adiatma, Y.D.**, Lindskog, A., Fravel, M.S., Crissey, J.T., Schwartz, M., White, G.A., Ahlberg, P., Reid, R.E.B., Lyons, T.W., Gill, B.C., Owens, J.D., Young, S.A., Oscillatory paleoredox conditions during the early Paleozoic. in review for *Earth and Planetary Science Letters*.
- Zhou, A., Zheng, W., Sahoo, S. K., Them II, T. R., Banerjee, Y., **Adiatma, Y. D.**, Owens, J. D., Mercury isotope record sedimentary host phase transition across the Ediacaran-Cambrian boundary, in review for *Earth and Planetary Science Letters*
- Wang, X-K., Liu, X-M., Husinec, A., Cao, C., Dera, G., **Adiatma, Y. D.**, Lithium isotope evidence for enhanced hydrothermal activity in the Jurassic, in review for *Earth and Planetary Science Letters*
- Li, K., Ma, A., Hu, X., Lu, C., **Adiatma, Y. D.**, Li, L., Koeshidayatullah, A., Zhang, He., Yang, D., Li, Z., Diagenetic overprinting of micrite stable isotopes: implication for paleo-reconstruction, in review for *Geochimica et Cosmochimica Acta*

in preparation **Adiatma, Y.D.**, Fantle, M.S., Haber, P.C., Griffith, E.M., Saltzman, M.R., Calcium isotope constraints on the role of diagenesis in an early Mississippian positive carbon isotope excursion. *in preparation*

Research Grants

2025	CRC Seed Grant Research Fund (\$70,000)
2020	Graduate Student Research Grants, Geological Society of America (\$2,800)
2020	Grants-in-Aid, American Association of Petroleum Geologist (\$2,500)
2017 - 2021	Friends of Orton Hall travel and research grant, The Ohio State University, School of Earth Sciences (varied from \$1,500 to \$5,000)
2013	L. Austin Weeks Grant American Association of Petroleum Geologist (\$500)

Teaching Experience

Lecturer (Instructor of Record), ESC3100 History of Earth System Fall 2024

Florida State University, Dept. Earth Ocean & Atmospheric Science

Designed and delivered lectures, developed course materials, and assessed student learning in an upper-level introductory class (47 students); Integrated active learning strategies to enhance student engagement and foster critical thinking.

Lecturer (Instructor of Record), ESC1000 Introductory Earth Science Fall 2024

Florida State University, Dept. Earth Ocean & Atmospheric Science

Designed and delivered lectures, developed course materials, and assessed student learning in a large introductory class (207 students); Integrated active learning strategies to enhance student engagement and foster critical thinking.

Guest Lecturer, ES2155/ENR2155 Energy and Environment Spring 2023

The Ohio State University, School of Earth Sciences

Delivered a lecture focused on Earth system dynamics and anthropogenic climate change; led and facilitated students in discussion that focused on connecting earth science concepts to current anthropogenic global warming.

Teaching Assistant, ES5189 Field Geology Summer 2023

The Ohio State University, School of Earth Sciences

Supported a six-week field camp in Utah, guided students through hands-on geological mapping and data collection exercises and coordinated field logistics to ensure safety and effective learning experiences for a group of students with diverse physical fitness, abilities, and academic backgrounds.

Teaching Assistant, ES1200 Introduction to Earth Science (Lab) Spring 2022

The Ohio State University, School of Earth Sciences

Taught lab sessions for introductory Earth science, incorporating practical exercises to solidify student comprehension of basic earth science concepts, and provided one-on-one support during office hours

Teaching Assistant, ES1122 Earth through Time (Lab) Fall 2020
The Ohio State University, School of Earth Sciences
Developed online lab materials and spearheaded efforts to transition the course to a virtual format during the pandemic, ensuring continuity in student learning, taught lab. sessions and provided academic support through virtual office hours, adapting teaching methods to remote learning.

Teaching Assistant, ES1121 Dynamic Earth (Lab) Spring 2019
The Ohio State University, School of Earth Sciences
Taught lab sessions for introductory Earth science, incorporating practical exercises to solidify student comprehension of basic earth science concepts and provided one-on-one support during office hours

Head Teaching Assistant, Sedimentology Spring – Fall 2013
Institut Teknologi Bandung, Geology Study Program
Led and managed a team of teaching assistants, developed lab materials, and organized geologic field trips for a class of over 80 students.

Teaching Assistant, Petroleum Geology and Well Logging & Petrophysics Fall 2013
Institut Teknologi Bandung, Geology Study Program
Taught lab sessions for Petroleum Geology and Well Logging & Petrophysics. Provided one-on-one support during office hours.

Undergraduate Student Mentoring

Mackenzie Fravel, Florida State University Summer 2024 - Fall 2025
Records of Early Ordovician Carbon Cycling in Deep Water Settings from Baltica

Josh Crassey, Florida State University Fall 2024 – Fall 2025
Records of Middle Ordovician Carbon Cycling in Deep Water Settings from Baltica

Blake Roberts, Florida State University Spring 2025 - Present

Lucas Carter, The Ohio State University Spring 2019 – Fall 2022
Thesis title: Lithofacies and Neodymium Isotope Stratigraphy of the Knox Unconformity in the Central Appalachian Basin

Graduate Student Mentoring

Gwen Barnes, Florida State University (PhD) Spring 2024 - Present
Thesis title: An investigation of the marine redox conditions, sedimentology, and biotic dynamics of the Early-Middle Ordovician Baltoscandian paleobasin.

Charlie Smith, Florida State University (MS) Fall 2024 - Present
Thesis title: Insights on redox conditions through the Great Ordovician Biodiversification Event from deep water facies of western Laurentia

Honors and Awards

2022	The Michael S. Johnson Outstanding Graduate Student Award, The Ohio State University, School of Earth Sciences
2010 – 2013	Dean's List, Institut Teknologi Bandung, Faculty of Earth Sciences and Technology

Technical Skills - Laboratory

Mass Spectrometry:

Thermal Ionization Mass Spectrometer (TIMS)

- Isotope analysis for Sr, Nd, Sm, and Ca isotopes
- Basic maintenance and troubleshooting

Multi-Collector Inductively Coupled Plasma Mass Spectrometer (MC ICP MS)

- Thallium isotope analysis

Inductively Coupled Plasma Mass Spectrometer (ICP MS)

- High-precision trace element analysis
- Method development for trace element analysis in geologic materials
- Basic maintenance and troubleshooting

Inductively Coupled Plasma Optical Emission Spectrometer (ICP OES)

- High-precision major and trace element analysis

Geochemistry

- Column chemistry (Sr, Li, Nd, Sm, Ca, U, Tl)
- Ultra-low blank sample processing for sub ppb elemental concentration analysis
- Iron speciation
- Chromium Reducible Sulphur Extraction
- UV Spectrophotometry
- Basic lab. maintenance

Technical Skills - Field Geology

Sedimentology and Stratigraphy

- High-resolution stratigraphic logging
- Facies analysis and sequence stratigraphy

Structural Geology and Tectonics

- Geologic Mapping
- Structural geology mapping and fault kinematic analysis

Technical Skills - Computational

Programming and Data Analytics, Python and Matlab

- Statistical analysis
- Data visualization using Matplotlib and Seaborn
- Statistical learning techniques (e.g., deep learning) using Sci-Kit Learn and PyTorch
- Geochemical Modeling (isotope mixing model, reservoir box model, diagenesis model)

Industry Standard Software (Petroleum Geology)

- Static reservoir modeling: Petrel
- Petrophysical analysis: Geolog, TechLog, and Interactive Petrophysics
- Seismic Stratigraphy: Kingdom

Field Work Experience

<i>Confusion and Pahrnagat Range, Utah and Nevada (USA), 7 days</i>	2026
<i>Trail Creek, Idaho (USA), 12 days</i>	2024
<i>Colliers town, Virginia (USA), 7 days</i>	2019
<i>Germany Valley, West Virginia (USA), 7 days</i>	2018
<i>Antelope Range, Nevada (USA), 7 days</i>	2018
<i>East River Mountain, West Virginia (USA), 7 days</i>	2017
<i>Tapin District, South Kalimantan (Indonesia), 35 days</i>	2013
<i>Nanga Mentebah, West Kalimantan (Indonesia), 10 days</i>	2013

Community Service and Leadership Roles

Reviewer	2023 - present
Geology, Science Advances, Geochimica et Cosmochimica Acta, Journal of Marine and Petroleum Geology, Sedimentary Geology, Chemical Geology, Journal of Sedimentary Research	

Session Organizer, GSA Southeastern Section Meeting 2025
T19. Co-evolution of Life and its Environment: From Biodiversification Events to Mass Extinction and Everything in Between

Organized and coordinated a scientific session for the Geological Society of America Southeastern Section Meeting, managing abstract review, speaker selection, and session logistics.

Earth Science Delegate, Council of Graduate Student 2022
The Ohio State University

Represented graduate students at the School of Earth Sciences at a university-level graduate student government. Together with other council members, played an active role in passing resolutions to improve graduate students' well-being. One of the resolutions we passed proposed an increase in university/employee health insurance contribution from 85% to 100%, which was later brought up to the university senate, approved, and implemented starting in the Fall 2023 semester.

Student Representative, Graduate Study Committee 2022
The Ohio State University, School of Earth Sciences

Represented graduate students in the department graduate study committee, hosted graduate student townhall meetings, conducted surveys to gauge graduate students' satisfaction and collected issues to bring up during townhall meetings, and together with other graduate study committee members, spearheaded efforts to improve graduate students' well-being (e.g., stipend adjustment, training for faculties and graduate student leaders).

Session Organizer, GSA Annual Meeting 2020
T63. The Ordovician Earth: New Insights to Environmental and Biotic Response in the Fossil and Rock Record

Organized and coordinated a scientific session for the Geological Society of America annual conference, managing abstract review, speaker selection, and session logistics.

Session Organizer, GSA Annual Meeting 2019
T116. The Ordovician Earth: Integrated Perspective on the Fossil and Rock Records

Organized and coordinated a scientific session for the Geological Society of America annual meeting in Phoenix, AZ, managing abstract review, speaker selection, and session logistics. Successfully managed session timeline and moderated discussions to ensure productive academic discourse.

President, AAPG Student Chapter 2012
Institut Teknologi Bandung

Led student organization operations including event planning, member engagement, budget management, and liaison activities with faculty and industry partners. Successfully secured full sponsorship from major oil companies for two geological field trips to Mahakam Delta and Northwest Java Basin, eliminating student costs and providing hands-on industry exposure to chapter members. Organized and coordinated multiple guest lecture series featuring petroleum industry professionals from various oil companies, enhancing students' understanding of current industry practices and career opportunities. Strengthened chapter's industry connections while significantly expanding educational opportunities for geology students.

Professional Associations

Indonesia Association of Geologists (IAGI)
Geological Society of America (GSA)
American Geophysical Union (AGU)

Professional Experience

Subsurface Geologist, Vico Indonesia (contract work via LAPI ITB) 2014 – 2016

Developed static reservoir models for the Semberah Field (Kutai Basin), using integrated geological and petrophysical data, performed petrophysical analysis and resource estimation using industry-standard software (Petrel and Geolog), collaborated with multidisciplinary teams to characterize reservoir properties and estimate remaining hydrocarbon resources.

Field Geologist, Noras Nusantara Summer 2012

Conducted comprehensive geological mapping of coal seam formations to support client exploration and development projects. Executed intensive 10-day field campaign in Nanga Mentebah, West Kalimantan, collecting geological data and samples across diverse terrain conditions. Prepared detailed technical reports documenting coal seam characteristics, geological interpretations, and resource assessments for client deliverables. Demonstrated proficiency in field data collection, geological logging, and translating complex geological findings into actionable recommendations for mining operations.

Field Engineer Intern, Schlumberger

Summer 2011

Gained hands-on experience in oilfield operations through direct participation in drilling and measurement activities at Limau Field South Sumatra. Assisted senior field engineers with tool lay down procedures and Measurement While Drilling (MWD) operations. Assisted in various sensor calibration processes to maintain accuracy of downhole measurement tools. Completed comprehensive intern project utilizing Petrel software for reservoir modeling and analysis.

Conference Presentations and Invited Talk

Invited Talk

- 2026 *Trace metal geochemistry constraints on paleoredox conditions during the early Paleozoic. "McConnel Club" Geological Survey of Canada-Calgary (GSCC), Natural Resources Canada (NRCan)*
- 2024 Introduction to chemostratigraphy. Indonesian Sedimentologist Forum
- 2023 Inorganic geochemistry and provenance studies. Pertamina University
- Selected conference abstracts from the past 4 years
- 2025 Smith, C., **Adiatma Y.D.**, Owens, J.D., Goldman, D., Leslie, S., Gill, B. C., Young, S. A., Insights on redox conditions through the great Ordovician Biodiversification Event from deep water facies of western Laurentia. Presented at the GSA Connects 2025 meeting in San Antonio, Texas
- 2025 Barnes, G. L., **Adiatma, Y. D.**, Lindskog, A., Owens, J. D., Young, S. A., An investigation of the local redox landscape of the Middle Ordovician Baltoscandian carbonate shelf using I/Ca records. Presented at the GSA Connects 2025 meeting in San Antonio, Texas
- 2025 Saltzman, M.R., Griffith, E.M., **Adiatma, Y.D.**, Haber, P., Fantle, M. S., Pairing Ca and C isotopes to disentangle diagenesis and carbon cycling during Paleozoic carbon isotope excursions. Presented at Goldschmidt Conference 2025 in Prague, Czech Republic
- 2024 **Adiatma, Y.D.**, Schwartz, M.J, Ahlberg, P., Owens, J.D. Young, S.A., Multiproxy chemostratigraphic constraints on paleoredox conditions during the Early to Middle Ordovician interval in the Baltic Basin. Presented at the GSA Connects 2024 meeting in Anaheim, California
- 2024 Goodin, J.T., Them, T.R., Caruthers, A.H., Hagen, A., Marroquin, S., McCabe, K., **Adiatma, Y.D.**, Grocke, D., Alexandre, J.T., Gill, B.C., Owens, J.D., A brief period of marine oxygenation during the End-Triassic Mass Extinction—a Thallium isotope modeling approach. Presented at the GSA Connects 2024 meeting in Anaheim, California

- 2024 Liu, X.M. Wang, X.K, Husinec, A., Cao, C., **Adiatma, Y.D.**, Tracing Ancient Hydrothermal Activity: Lithium Isotope Insights into the Jurassic Adriatic Platform. Presented at Goldschmidt Conference 2024 in Chicago, Illinois
- 2024 Saltzman, M.R., Griffith, E.M., **Adiatma, Y.D.**, Al-Musawi, M., The Cambrian SPICE event: perturbation of global carbon cycle or global diagenesis? Presented at Goldschmidt Conference 2024 in Chicago, Illinois
- 2023 **Adiatma, Y.D.**, Saltzman, M.R., Griffith, E.M., Haber, P.C., Braun, M.G., Edwards, C.T., Diamond, C.W., Calcium isotope constraints on the origin of the Mid-Darriwilian Carbon Isotope Excursion (MDICE). Presented at the GSA Connects 2023 meeting in Pittsburgh, Pennsylvania
- 2022 **Adiatma, Y.D.**, Saltzman, M.R., Wang, X.K., Liu, X.M., Constraining changes in silicate weathering during the Early Ordovician using lithium isotope chemostratigraphy. Presented at the GSA Connects 2024 meeting in Denver, Colorado
- 2022 Haber, P.C., Saltzman, M.R., Griffith, E.M., **Adiatma, Y.D.**, Bergmann, K.D., Anderson, N.T., The application of calcium isotopes to understand the effect of diagenesis on carbon isotope trends in ancient carbonate: an example from the Early Mississippian. Presented at the GSA Connects 2024 meeting in Denver, Colorado
- 2022 **Adiatma, Y.D.**, Saltzman, M.R., Griffith, E.M., Haber, P.C., Edwards, C.T., Diamond, C.W., Calcium isotope constraints on diagenetic effects in carbon isotope ($\delta^{13}\text{C}$) data: a case study from Middle Ordovician Carbonate Strata at Meiklejohn Peak, Nevada. Presented at the AGU 2023 Fall meeting in Chicago, Illinois
- 2022 **Adiatma, Y.D.**, Saltzman, M.R., Griffith, E.M., O'Neill, B.E., Chemostratigraphic correlation of a Darriwilian unconformity in the Appalachian Basin. Presented at the GSA North-Central and Southeastern 2024 Section meeting in Cincinnati, Ohio
- 2022 Carter, L.C., Saltzman, M.R., Griffith, E.M., **Adiatma, Y.D.**, Conwell, C.T., Lithofacies and Nd isotope stratigraphy of the Knox Unconformity in the Central Appalachian Basin. Presented at the GSA North-Central and Southeastern 2024 Section meeting in Cincinnati, Ohio
- 2022 Haber, P.C., Saltzman, M.R., Griffith, E.M., **Adiatma, Y.D.**, Early Mississippian calcium isotope stratigraphy and implications for conditions of carbonate deposition. Presented at the GSA North-Central and Southeastern 2024 Section meeting in Cincinnati, Ohio

References

Matthew R. Saltzman (PhD advisor)
The Ohio State University
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Florida State University
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Xiaoming Liu (Collaborator)
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